

INDIANAPOLIS EXECUTIVE, IN, USA

WMO: 720575

Lat: 40.031N

Lon: 86.251W

Elev: 281

StdP: 97.99

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00173

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.0	-22.3	0.5	-17.6	-19.4	0.7	-14.2	11.3	2.8	10.6	-0.4	2.3	290	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.4	23.4	31.1	23.1	29.8	22.5	25.5	30.1	24.5	29.0	23.6	27.9	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.6	28.2	23.0	18.3	27.0	22.4	17.7	26.2	79.7	30.1	75.6	29.1	72.0	28.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.2	8.7	7.8	-22.9	34.4	3.6	2.2	-25.5	36.0	-27.5	37.2	-29.5	38.4	-32.1	40.0		
			-23.1	26.6	3.5	1.3	-25.7	27.5	-27.7	28.3	-29.7	29.0	-32.2	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-3.5	-1.0	5.3	11.2	17.7	22.1	23.0	21.5	18.7	12.1	5.1	0.1
	DBStd	10.55	6.78	6.78	6.21	5.08	5.00	3.21	3.20	2.72	3.93	5.07	5.46	5.83
	HDD10.0	1470	421	312	174	46	4	0	0	0	0	36	166	311
	HDD18.3	3144	678	541	407	220	74	7	3	5	42	204	398	566
	CDD10.0	1863	1	4	28	81	243	363	402	358	263	100	17	3
	CDD18.3	496	0	0	1	5	55	121	146	104	54	9	0	0
	CDH23.3	4355	0	0	11	47	479	979	1335	902	516	85	1	0
	CDH26.7	1336	0	0	1	3	130	313	474	245	158	12	0	0
Wind	WSAvg	3.3	4.2	4.1	4.0	4.1	3.4	3.0	2.2	1.8	2.1	3.1	3.6	3.9
Precipitation	PrecAvg	1016	65	57	85	99	107	98	112	84	78	71	87	75
	PrecMax	1347	245	151	214	208	270	214	354	200	221	201	257	194
	PrecMin	738	7	9	16	26	28	7	14	4	4	5	7	12
	PrecStd	171	46	30	43	44	54	45	74	43	53	39	50	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	18.2	26.0	27.2	32.2	32.9	36.0	32.4	32.9	28.0	22.4	17.3
		MCWB	13.1	13.3	18.2	16.6	21.9	22.7	25.1	23.5	22.5	20.3	14.3	14.2
	2%	DB	11.8	14.9	21.2	24.9	29.8	31.4	32.7	31.0	30.5	25.9	19.0	14.1
		MCWB	10.5	12.3	15.2	16.2	20.4	22.5	24.3	23.4	22.2	18.2	13.8	12.8
	5%	DB	8.9	12.2	17.7	22.5	27.9	30.1	31.1	29.2	28.4	23.2	16.2	11.1
		MCWB	7.4	10.1	13.4	14.8	19.5	22.0	24.1	22.6	21.3	17.1	12.2	9.5
	10%	DB	5.7	8.9	15.2	20.0	26.3	28.4	29.2	27.8	26.5	20.9	13.7	8.0
		MCWB	4.1	6.7	11.1	13.9	19.2	21.1	23.3	21.7	19.6	16.0	10.0	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.0	18.5	19.0	23.2	25.5	26.9	26.3	24.7	22.5	17.2	15.1
		MCDB	15.0	16.9	25.7	24.3	28.6	30.7	31.8	29.8	29.5	26.4	21.0	16.5
	2%	WB	10.4	12.5	16.0	17.4	21.9	24.2	25.9	24.8	23.6	20.4	14.9	12.7
		MCDB	11.3	13.9	19.7	21.8	27.3	29.2	30.8	28.9	28.4	23.2	17.5	14.0
	5%	WB	7.6	10.1	14.1	16.2	21.0	23.2	25.1	23.8	22.4	18.7	12.6	9.4
		MCDB	8.5	12.4	17.5	20.2	26.0	27.9	29.5	27.7	26.7	21.1	16.0	10.8
	10%	WB	4.1	6.6	11.9	15.1	20.1	22.3	24.0	22.7	21.4	16.8	10.3	6.6
		MCDB	5.9	8.7	14.8	19.0	24.8	26.8	28.2	26.4	24.9	20.7	13.3	7.9
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.7	9.5	11.3	10.8	10.3	10.5	11.5	12.2	11.6	9.6	7.3
		MCDBR	10.6	12.0	13.0	14.5	12.5	12.2	12.0	12.7	14.0	14.2	13.3	10.5
		MCWBR	9.3	10.0	8.4	8.1	5.7	5.4	5.2	6.4	6.5	7.8	8.9	9.3
	5% WB	MCDBR	10.2	11.1	11.6	11.3	10.8	10.3	10.6	10.4	11.5	10.8	11.9	10.1
		MCWBR	9.7	9.8	7.9	7.7	5.6	5.3	5.3	5.9	6.1	6.8	8.9	9.4
Clear-Sky Solar Irradiance	taub	0.300	0.324	0.345	0.403	0.439	0.435	0.451	0.450	0.413	0.358	0.319	0.299	
	taud	2.424	2.337	2.369	2.213	2.164	2.215	2.174	2.163	2.252	2.403	2.461	2.449	
	Ebn at Noon	866	889	909	870	844	845	827	817	825	841	837	837	
	Edh at Noon	81	102	110	138	148	141	146	144	123	95	78	74	
All-Sky Solar Radiation	RadAvg	1.74	2.57	3.64	4.73	5.41	6.15	6.05	5.47	4.51	3.13	2.07	1.45	
	RadStd	0.18	0.28	0.30	0.41	0.49	0.38	0.35	0.32	0.36	0.29	0.23	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+97	N/A

Nomenclature: See separate page